

Digital OPD Desk – Product Data Document

Document Version: 1.0
Last Updated: 2026-09-03
Product: Digital OPD Desk (by Medcy Health Tech)
Purpose: Reference document for MOU drafting with hospitals and clinics.

 **This is a living document.** Must be updated whenever the platform is updated with new features or changes.

1. What is the Digital OPD Desk?

The Digital OPD Desk is a cloud-based software platform that digitizes the complete outpatient department workflow for hospitals and clinics. It replaces paper-based token systems, manual patient registers, and fragmented record-keeping with a single unified system.

In simple terms: From the moment a patient walks into the hospital to the moment they leave with a prescription – everything is digital.

2. Who Uses It?

User	What They Do on the Platform
Receptionist	Registers patients, schedules appointments, manages the token queue, uploads documents
Doctor	Views patient queue, records clinical notes, writes prescriptions, tracks vitals, manages treatments
Nurse	Records patient vitals (BP, temperature, SpO2, etc.), uploads reports
Clinic Admin	Manages staff, views dashboard & analytics, configures room/bed setup
Patient	Views their queue position, upcoming appointments, reports, and prescriptions via a self-service portal

3. Modules & Features

3.1 Patient Registration

When a new patient visits the hospital, the receptionist registers them in the system.

Information captured:

- Full name (*required*)
- Mobile number (*required, unique per clinic*)
- UHID – Unique Hospital ID (*auto-generated by the system*)
- Gender, Date of Birth, Age, Blood Group
- Marital Status
- Aadhar Number (Government ID)
- Full address (House, Street, Area, City, District, State, PIN Code)
- Emergency contact (Name, Phone, Relation)
- Assigned doctor
- Referral doctor (if any)

Automatic actions on registration:

- A unique UHID is assigned to the patient.
- A 4-digit PIN is generated for the patient to access the Patient Portal.

3.2 Token & Queue Management (QMS)

This is the core of the OPD desk. When a patient checks in, they receive a **token number** and are placed in the doctor's queue.

How it works:

1. Patient arrives → Receptionist checks them in → Token number is generated.
2. Patient enters the **waiting queue** for their assigned doctor.
3. Doctor's dashboard shows the live queue with token numbers.
4. Doctor **calls** the next patient → patient status changes to "Called".
5. Consultation begins → status changes to "In Consultation".
6. Consultation ends → status changes to "Completed".

Queue controls available to staff:

Action	What It Does
Call	Calls the next patient in line
Skip	Skips a patient (e.g., not present) – can be re-queued later
Resume	Puts a skipped or no-show patient back into the queue

Action	What It Does
Complete	Marks the consultation as done
Cancel	Removes the patient from the queue
No Show	Marks patient as absent

Queue statuses:

BOOKED → ARRIVED → WAITING → CALLED → IN CONSULTATION → COMPLETED

Patients can also be marked as SKIPPED, CANCELLED, or NO SHOW at various stages.

Estimated wait time:

- The system calculates how long a patient will likely wait based on the doctor's average consultation speed and the number of patients ahead in the queue.

Walk-in patients:

- Walk-in patients can be registered and added to the queue in a single step – the system automatically finds the next available slot for the doctor.

Automatic cleanup:

- Appointments from previous days that were never completed are automatically closed by the system (marked as "Completed" if clinical data exists, or "No Show" if not).

Patient notifications:

- When it's almost a patient's turn (configurable – default: 2 patients ahead), the system can send a WhatsApp/SMS notification.

3.3 Appointment & Doctor Scheduling

Doctor Schedule Setup:

Clinic admin or doctor can set up weekly schedules:

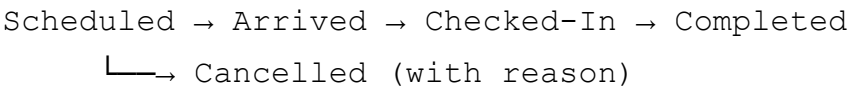
- Working days (Monday to Sunday)
- Working hours (e.g., 9:00 AM to 5:00 PM)
- Break times (e.g., 1:00 PM to 2:00 PM)
- Slot duration (e.g., 15 minutes per patient)
- How many patients can be booked per slot (slot capacity)

Once a schedule is saved, the system **automatically generates available slots for the next 30 days**.

Appointment types supported:

Type	Description
Consultation	Regular OPD visit
Follow-up	Return visit
Procedure	Medical procedure
Emergency	Emergency visit
Scan	Diagnostic scan
Surgery	Surgical procedure
Camp	Health camp / screening

Appointment status flow:



Double-booking prevention:

- The system prevents two patients from being booked in the same time slot with the same doctor (unless slot capacity allows multiple patients).

3.4 Clinical Notes & Documentation

Doctors can record clinical notes for each patient visit.

SOAP Notes (Structured):

Section	What It Captures
Subjective	Patient's complaint and history ("Patient reports headache for 3 days")
Objective	Doctor's examination findings ("BP: 130/85, Temp: 99°F")
Assessment	Diagnosis ("Tension-type headache")
Plan	Treatment plan ("Prescribe paracetamol, follow-up in 1 week")

Quick Notes:

- Doctors can also write free-text quick notes for simple observations.

Important: Clinical notes support soft-delete — deleted notes are hidden but retained in the system for legal and compliance purposes.

3.5 Prescriptions

Doctors can write digital prescriptions with the following details per medication:

Field	Example
Medication Name	Paracetamol 500mg
Dosage	1 tablet
Frequency	Twice daily
Duration	5 days
Special Instructions	Take after meals

- Multiple medications can be grouped into a single prescription.
- Prescriptions can be marked as Active or Cancelled.
- Patients can view their prescriptions on the Patient Portal.

3.6 Patient Vitals

Nurses or doctors can record vital signs for each patient visit:

Vital	Example
Blood Pressure	120/80 mmHg
SpO2 (Oxygen)	98%
Temperature	98.6°F
Weight	65 kg
Height	170 cm
Pulse Rate	72 bpm

- Vitals can be linked to a specific appointment.
- Vitals update in **real-time** on connected dashboards.

3.7 Allergies & Medical History

Allergies:

Field	Example
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Field	Example
Allergy Name	Penicillin
Severity	Low / Medium / High / Critical
Reaction	Skin rash

Medical History:

Field	Example
Condition	Type 2 Diabetes
Diagnosis Date	2020-03-15
Status	Active / Resolved

3.8 Treatment Plans

For ongoing treatments (e.g., physiotherapy sessions, dental procedures):

Field	Example
Treatment Name	Root Canal Treatment
Description	Multi-session root canal for lower molar
Status	Planned / In Progress / Completed / Cancelled
Cost	₹5,000

- Multiple sessions can be linked to one master treatment plan.

3.9 Patient Timeline

Everything recorded for a patient – appointments, clinical notes, prescriptions, vitals, allergies, medical history, reports, and treatments – is shown in a **single chronological timeline**.

This gives doctors a complete picture of the patient's journey at a glance.

3.10 Document & Report Management

Staff can upload patient documents (lab reports, scan images, discharge summaries, etc.) to the system.

How it works:

1. Staff uploads a document (max 25 MB per file).
2. If the patient is known, the document is linked directly to the patient.
3. If the patient is unknown (e.g., report arrived before registration), the document is placed in an **"Unassigned" triage queue**.
4. Later, staff can link the unassigned document to the correct patient – or create a new patient and link the document in one step.

Document types: Lab reports, prescriptions, discharge summaries, consultation notes, imaging, and any custom type.

All documents are stored securely in the cloud with time-limited access links.

3.11 Room & Bed Management (IPD)

For hospitals that also manage inpatient admissions:

Setup:

- Define **room categories** (e.g., ICU, General Ward, VIP Suite, Maternity Ward) with daily billing rates.
- Add **rooms** under each category (e.g., Room 101, Room 102).
- Add **beds** under each room (e.g., Bed A, Bed B).

Admission flow:

1. Patient is admitted → assigned to a bed → bed status changes to "Occupied".
2. Daily rate at the time of admission is locked in for billing accuracy.
3. Patient can be **transferred** to a different bed/room if needed.
4. On **discharge**, the bed is automatically freed up.

Bed statuses: Available, Occupied, Maintenance, Reserved.

Safety: The system prevents double-booking – a bed that is already occupied cannot be assigned to another patient.

3.12 Patient Portal (Patient Self-Service)

Patients can log in using their **mobile number + 4-digit PIN** to access:

Feature	Description
My Queue Position	See current position in the doctor's queue and estimated wait time

Feature	Description
My Appointments	View upcoming and past appointments
My Reports	Access and download lab reports, prescriptions, and other documents
My Prescriptions	View active and past prescriptions
My Vitals	View recorded vital signs history
My Profile	View personal details (name, UHID, blood group, etc.)

Security:

- 3 wrong PIN attempts → account locked for 20 minutes.
- Session expires after 1 hour for safety.

3.13 Dashboard & Analytics

For Clinic Staff / Admin:

Metric	Description
Patients served today	How many consultations were completed
Average consultation time	How long each doctor takes per patient on average
No-show rate	Percentage of patients who didn't turn up
Walk-in vs. Scheduled ratio	How many patients were walk-ins vs. pre-booked
Peak hours	Busiest hours of the day

Exportable: Queue data can be downloaded as a CSV/Excel file for offline reporting.

3.14 Staff Management

- Clinic admin can add staff members (doctors, receptionists, nurses) and assign them roles.
- Each staff member can be assigned to one or more clinic branches.
- Access is strictly role-based – a receptionist cannot access clinical notes, and a nurse cannot write prescriptions.

4. Data Security & Privacy

What We Protect	How We Protect It
Patient personal data	Encrypted at rest in the database
Sensitive medical information	AES-256 encryption (industry standard)
Patient login PIN	Hashed (one-way encryption – even we cannot see it)
Staff passwords	Hashed (one-way encryption)
Uploaded documents	Stored in encrypted cloud storage (AWS)
All communications	Encrypted in transit (HTTPS)

Data isolation:

- Each hospital/clinic's data is **completely isolated** from other hospitals on the platform. No clinic can access another clinic's data.

Audit trail:

- Every access to patient data is logged – who accessed what, when, and why.
- Every queue status change is logged for accountability.

Compliance:

- The platform follows HIPAA-style data handling practices.
- Row-Level Security is enforced at the database level.

5. How Hospitals/Clinics Access the Platform

Aspect	Detail
Access	Via web browser (works on desktop, tablet, and mobile)
Hosting	Cloud-hosted (no hardware installation needed at the hospital)
Data location	India
Uptime commitment	99.5% availability
Support	Dedicated support as per MOU terms

6. Summary of Data Handled by the Platform

Data Category	Examples
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Data Category	Examples
Patient demographics	Name, mobile, age, gender, blood group, address, Aadhar, emergency contacts
Clinical records	SOAP notes, prescriptions, vitals, allergies, medical history, treatment plans
Appointment records	Date, time, doctor, type, status, visit reason, referral info
Queue data	Token number, queue status, wait time, call time, consultation time
Documents	Lab reports, prescriptions, discharge summaries, scan images
Room/bed records	Room/bed assignments, admission/discharge dates, daily rates
Staff records	Name, email, role, assigned clinic(s)
Audit logs	Data access history, queue transition history, login activity

7. Change Log

Update this section every time the platform is updated with new features or changes.

Date	Version	What Changed
2026-09-03	1.0	Initial document – covers all existing OPD desk modules

End of Document